

Android App Development– Task 1

Build a Simple "Notes" App (Single Activity, Kotlin + Jetpack Compose)

Objective:

Get comfortable with Android project setup, Kotlin basics, Jetpack Compose UI, and local persistence (Room or simple file). You will create a single-screen notes app that lists notes and allows adding a new note.

Why this task?

It covers modern Android toolchain and UI toolkit (Compose), and gives a tiny but complete app that can be extended in later tasks.

Skills you'll gain

- Configure Android Studio and create a Kotlin Compose project.
- Build composable UIs (TextField, Buttons, LazyColumn).
- Use ViewModel and simple local persistence (Room or in-memory for Task-1).
- Run and test on emulator or physical device.

What you'll build

A minimal Notes app where:

- The main screen shows existing notes (title + snippet).
- A top bar and floating action button open a dialog to add a new note.
- Notes persist in-memory (for Task-1) or in Room (optional bonus).

Step-by-step

1. Install Android Studio ([follow Android Developers get-started](#)).
2. Create a new project → *Empty Compose Activity* (Kotlin).
3. Create a Note data class and a NotesViewModel to hold a list.
4. Build UI with Scaffold, TopAppBar, LazyColumn, and FloatingActionButton.
5. Implement adding a note via a Compose dialog.
6. Run on an emulator and capture screenshots.

Starter code (Kotlin + Jetpack Compose)

Note.kt

```
kotlin

data class Note(
    val id: Long = System.currentTimeMillis(),
    val title: String,
    val body: String
)
```

NotesViewModel.kt

```
kotlin

class NotesViewModel : ViewModel() {
    private val _notes = MutableStateFlow<List<Note>>(emptyList())
    val notes: StateFlow<List<Note>> = _notes

    fun addNote(title: String, body: String) {
        val new = Note(title = title, body = body)
        _notes.value = listOf(new) + _notes.value
    }
}
```

MainActivity.kt (Compose parts only)

```
kotlin

@Composable
fun NotesApp(viewModel: NotesViewModel = viewModel()) {
    val notes by viewModel.notes.collectAsState()
    Scaffold(
        topBar = { TopAppBar(title = { Text("My Notes") }) },
        floatingActionButton = {
            FloatingActionButton(onClick = { /* open dialog to add note */ }) {
                Icon(Icons.Default.Add, contentDescription = "Add")
            }
        }
    ) { padding ->
        LazyColumn(contentPadding = padding) {
            items(notes) { note ->
                Card(modifier = Modifier
                    .fillMaxWidth()
                    .padding(8.dp)) {
                    Column(modifier = Modifier.padding(12.dp)) {
                        Text(note.title, fontWeight = FontWeight.Bold)
                        Spacer(modifier = Modifier.height(6.dp))
                        Text(note.body, maxLines = 2, overflow = TextOverflow.Ellipsis)
                    }
                }
            }
        }
    }
}
```

Deliverables

1. GitHub repo with project and README.
2. APK or Play-Internal test link (optional).
3. Screenshots of app running on emulator.
4. Short write-up: how to run and extend the app (Room integration, search, delete).

Support Resources

Topic: Notes App Using Kotlin + Jetpack Compose

▶ YouTube Video Tutorials (Very Beginner-Friendly)

Jetpack Compose Basics

1. Build Your First Android App with Jetpack Compose
<https://www.youtube.com/watch?v=czk-VYbNfAI>
2. Jetpack Compose Beginner Tutorial (Hands-on Build)
<https://www.youtube.com/watch?v=jc3ZAs17uL8>
3. Jetpack Compose Full Beginner Course (freeCodeCamp — 5 Hours)
<https://www.youtube.com/watch?v=RebA5J-rlwg>
4. Notes App in Jetpack Compose (Real App)
<https://www.youtube.com/watch?v=BBWyXo-3JGQ>
5. Kotlin Full Course — Official JetBrains Tutorial
<https://www.youtube.com/watch?v=F9UC9DY-vIU>

Free Study Material & Docs

- Android Developers: Jetpack Compose Intro
<https://developer.android.com/jetpack/compose>
- Kotlin Language Documentation
<https://kotlinlang.org/docs/home.html>
- Android Studio User Guide
<https://developer.android.com/studio/intro>